SYNOPSIS

FOR

AMAZON PRODUCT ANALYSIS DASHBOARD

In partial fulfilment of the requirement for to award of the diploma of

Computer Engineering

Submitted by:

Sumit(230040824011)

Sumit

Vishvash (230040824013)

Vishvash

Submitted to:

Mr. Rajesh Kumar

Guru Daksh Government Polytechnic Hisar-125001

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INTRODUCTION:

With the rapid growth of e-commerce platforms such as Amazon, a large volume of product data is generated for every search result. This data includes product names, prices, customer ratings, number of reviews, and product links. Analyzing this raw data manually is difficult and time-consuming due to inconsistency and lack of structure in the dataset.

The project **"Amazon Product Analysis Dashboard"** is a desktop-based application developed using Python to analyze Amazon product search result data stored in Excel format. The system cleans and processes the raw data, performs statistical analysis, identifies top products and value-for-money products, visualizes important trends using graphs, and generates a complete PDF analysis report.

The project also provides a user-friendly graphical interface with multiple themes, interactive product links, and resizable visualization panels. This application helps users and analysts make informed decisions by presenting meaningful insights in a clear and organized manner.

OBJECTIVEs:

The main objectives of the project are:

- To analyze Amazon product search result data efficiently
- To clean and preprocess raw product data for accuracy and consistency
- To identify **Top 10 Products** based on ratings and reviews
- To identify **Decent Deals** (mid-price products with high ratings)
- To visualize data using graphs such as price distribution and rating analysis
- To generate a **complete PDF report** containing tables and graphs
- To develop an interactive and theme-based graphical user interface

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METHODOLOGY / PLANNING OF WORK:

The development of this project follows a systematic and analytical approach. The methodology includes the following steps:

1. **Data Collection**
   Amazon product search result data is collected in Excel format.

2. **Data Cleaning**
   The raw data is cleaned by removing missing values and normalizing product price, rating, and review count fields.

3. **Data Analysis**
   Statistical analysis is performed to compute summary statistics and rank products based on ratings and reviews.

4. **Product Classification**

   - Top 10 products are identified using rating and review metrics

   - Decent Deals products are identified using mid-price range and rating threshold

5. **Data Visualization**
   Graphs such as price distribution, rating vs price, and reviews vs rating are generated using Matplotlib.

6. **GUI Development**
   A Tkinter-based desktop GUI is developed with resizable graph panels, clickable product links, and multiple color themes.

7. **Report Generation**
   A detailed PDF report is generated using ReportLab, including tables, statistics, and visual graphs.

• FACILITIES REQUIRED FOR PROPOSED WORK

Software Requirements

• Python 3.x

• Anaconda Distribution / Python Interpreter

• Pandas Library

• NumPy Library

• Matplotlib Library

• Tkinter (GUI Development)

• ReportLab (PDF Generation)

• Microsoft Excel

Hardware Requirements

• Personal Computer / Laptop

• Minimum 4 GB RAM

• Intel / AMD Processor

APPLICATIONS

- E-commerce product comparison
- Market research and analysis
- Business intelligence demonstrations
- Academic and mini-project work
- Decision support systems

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2. Python Official Documentation – https://www.python.org
3. Tkinter GUI Programming – Python Documentation
4. Pandas Documentation – https://pandas.pydata.org
5. Matplotlib Documentation – https://matplotlib.org